ml-models-lab

Five ML models built from scratch for a B2B distributor setting - method, one improvement, honest measured result.

All data synthetic and seeded; numbers are exactly what the code prints at the default seed. No euro claims.

`demand-forecast-net`

Global NB MLP (DeepAR-lite) -> intermittent routing + Croston states, inverse-scale loss

Measured: MASE 0.987 / RMSSE 0.948 vs seasonal-naive 1.080/1.062 and Holt-Winters 1.101/1.019

`sku-text-classifier`

fastText hashed char n-grams -> regex part-number split + inverse-frequency class weights

Measured: macro-F1 0.963 vs majority-class baseline 0.013

`order-anomaly-ae`

Reconstruction-error AE -> per-feature explanations + clean-split threshold; PCA baseline

Measured: AE PR-AUC 0.963 vs PCA 0.951 - narrow win; PCA recommended as the first reach

`churn-rfm-predictor`

From-scratch numpy logistic regression -> frequency-slope feature + Platt calibration

Measured: PR-AUC 0.653 vs recency 0.361 / prevalence 0.150; ECE 0.197 -> 0.021 after Platt

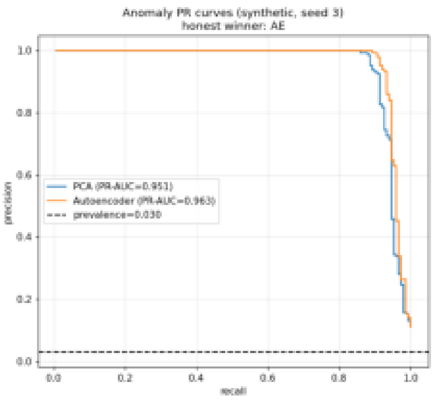
`price-elasticity-regressor`

From-scratch ridge/lasso, log-log elasticity -> hierarchical shrinkage + endogeneity control

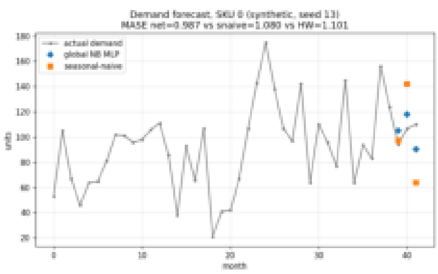
Measured: naive-OLS bias +1.52 -> +0.03 controlled; profit regret 0.89% vs analytic optimum

Honesty highlights

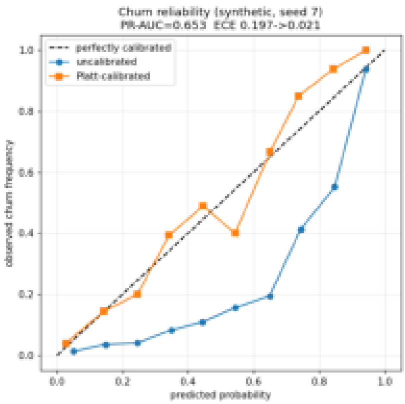
- PCA over the autoencoder: the AE wins only narrowly (PR-AUC 0.963 vs 0.951), so the simpler PCA-SVD baseline stays in the report and is the recommended first reach.
- Naive-baseline transparency: seasonal-naive and Holt-Winters sit in the forecasting headline table; recency and prevalence beside churn; majority-class beside macro-F1.



AE vs PCA precision-recall curves



Forecasts vs the naive baselines



Platt calibration: ECE 0.197 -> 0.021

Reproduce it

```
pip install -r requirements.txt       # numpy / scipy / pandas / matplotlib / torch
python -m mllab.<model>               # train + print metrics + save the labelled figure
python -m ruff check .                # lint gate (clean)
python -m pytest -q                   # 20 deterministic / baseline / invariant tests
python tools/make_onepager.py         # rebuild this PDF from the repo figures
```

Deliverable of the ml-models-lab repo - full specs in docs/METHODOLOGY.md, framing in docs/BUSINESS_CASE.md.
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